

Replication Plan

Quantitative Relationship Between Polarization Differences and the Zone-Averaged
Shift Photocurrent

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Paper Overview

This paper establishes a quantitative connection between the bulk photovoltaic (shift current) response and the difference in electronic polarization between valence and conduction bands. Using tight-binding models on 1D and 2D lattices (Rice–Mele model, Haldane model variants), the authors derive a zone-averaged shift-current expression relating the shift vector integral to polarization differences. The work is entirely analytic/numerical, with all model parameters specified.

Detailed Workflow

Phase 1: Tight-Binding Model Construction

1.1 1D Rice–Mele model.

- Two-site unit cell with alternating hopping t_1, t_2 and on-site energies $\pm\Delta$.
- Construct the 2×2 Bloch Hamiltonian $H(k)$.

1.2 2D models (Haldane-type, if applicable).

- Construct the Bloch Hamiltonian on the honeycomb lattice with next-nearest-neighbor hopping and complex phases.

Phase 2: Band Structure and Berry Phase Calculations

2.1 Diagonalize $H(k)$ across the Brillouin zone on a dense k -mesh.

2.2 Compute Berry connection $\mathcal{A}_n(k) = i\langle u_n(k) | \nabla_k | u_n(k) \rangle$ for each band n .

2.3 Compute Berry phase / polarization $P_n = \frac{1}{2\pi} \oint \mathcal{A}_n(k) dk$ via the discretized Wilson loop.

2.4 Compute polarization difference $\Delta P = P_{\text{cond}} - P_{\text{val}}$.

Phase 3: Shift Current Calculation

3.1 Compute the shift vector $R_{cv}(k)$ for each k -point:

$$R_{cv}(k) = \frac{\partial \phi_{cv}(k)}{\partial k} - [\mathcal{A}_c(k) - \mathcal{A}_v(k)]$$

where ϕ_{cv} is the phase of the interband matrix element.

3.2 Compute the shift current conductivity $\sigma^{\text{shift}}(\omega)$ by integrating $R_{cv}(k)$ weighted by the joint density of states over the BZ.

3.3 Compute the zone-averaged shift current and verify the relation to ΔP .

Phase 4: Parameter Sweeps and Figure Reproduction

4.1 Sweep model parameters (t_1/t_2 , Δ , etc.) as shown in the paper.

4.2 Plot shift current vs. polarization difference.

4.3 Reproduce all figures and tables from the paper.

4.4 Verify analytical limiting cases.

Datasets

No external datasets; all data is generated from model Hamiltonians.

Dataset / Source	Type	Location / Notes
Model parameters	Synthetic	Specified in paper text and figure captions
Band structure data	Computed	Generated from Hamiltonian diagonalization
Paper figures (reference)	Reference	For quantitative comparison of reproduced results

Models, Tools, and Software

Programming Languages

- **Python 3.8+** or **Julia 1.6+** — for numerical implementation.

Libraries and Frameworks

Tool / Library	Version	Purpose / URL
NumPy	≥ 1.21	Matrix diagonalization, array operations
SciPy	≥ 1.7	Numerical integration, interpolation
Matplotlib	≥ 3.4	Plotting band structures, shift current spectra
PythTB	latest	Tight-binding model construction, Berry phase https://www.physics.rutgers.edu/pythtb/
Z2Pack	latest	Wilson loop and topological invariant calculations https://z2pack.greschd.ch/
WannierTools	latest	Optional: Wannier-based Berry phase calculations https://www.wanniertools.org/

Hardware Requirements

- Any modern laptop; 1D models are computationally trivial; 2D models require a dense k -mesh but remain fast.